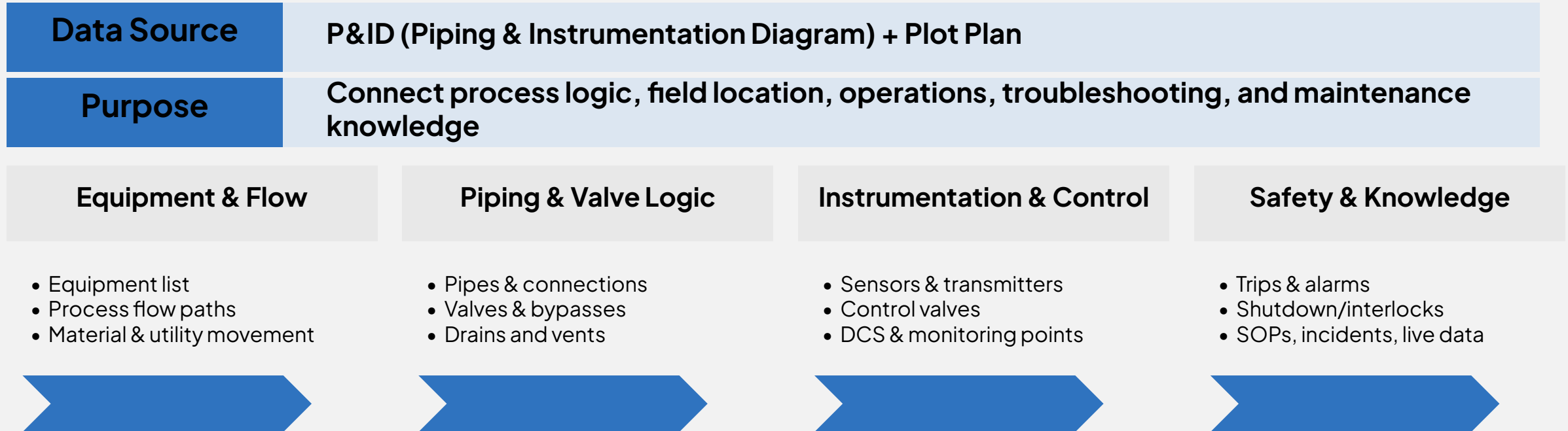
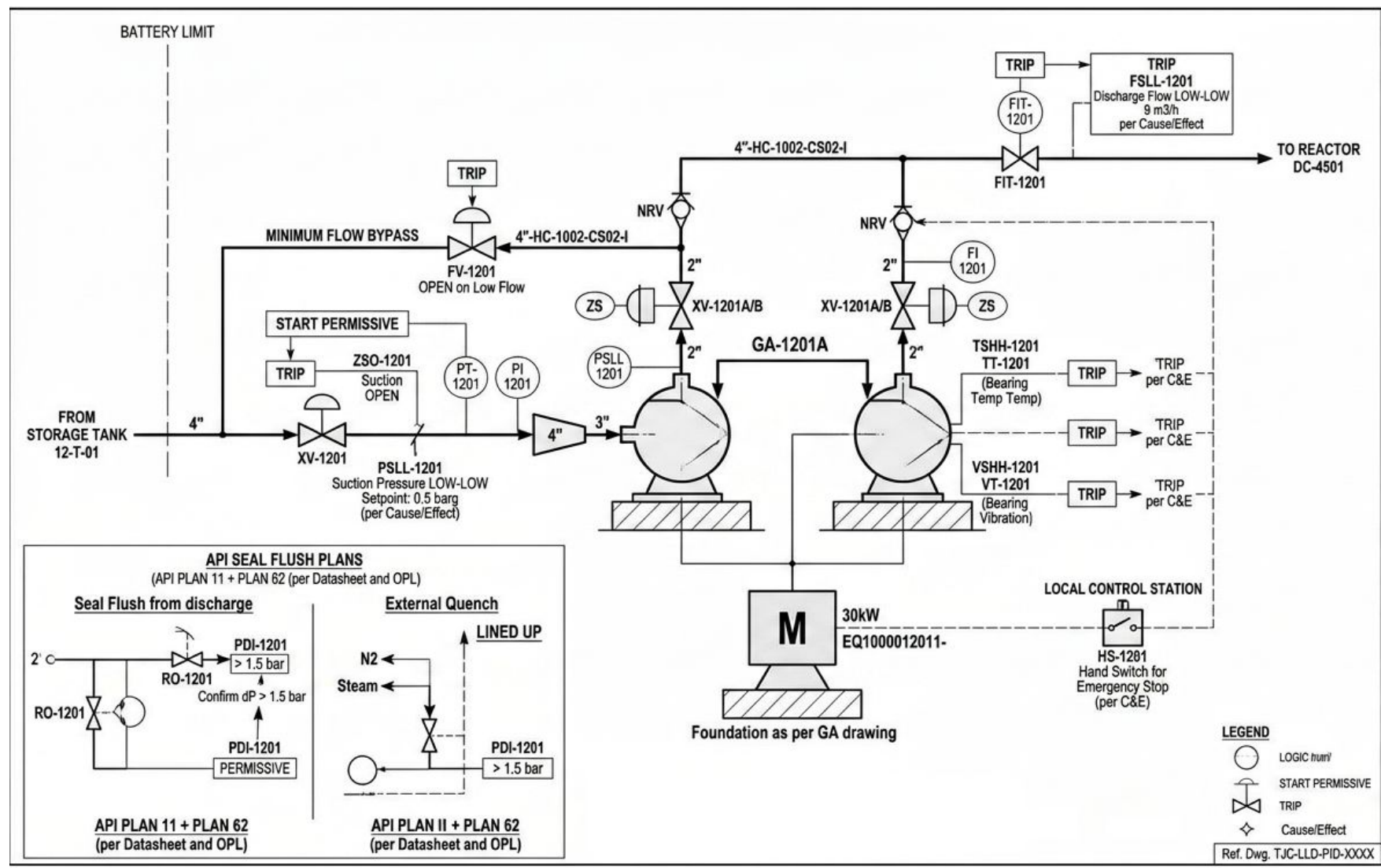


P&ID & Plot Plan Overview



**P&ID explains how the system works. Plot Plan shows where equipment is located.
Together they create operational context for decision-making.**



NOTES:
NOTE 1: This is sample data provided for CALIBER purposes only;
NOTE 2: Refer to Datasheet TJC-LLD-DS-GA-1201A for design parameters;
NOTE 3: See SEQ-1201 logic diagram.

What it shows

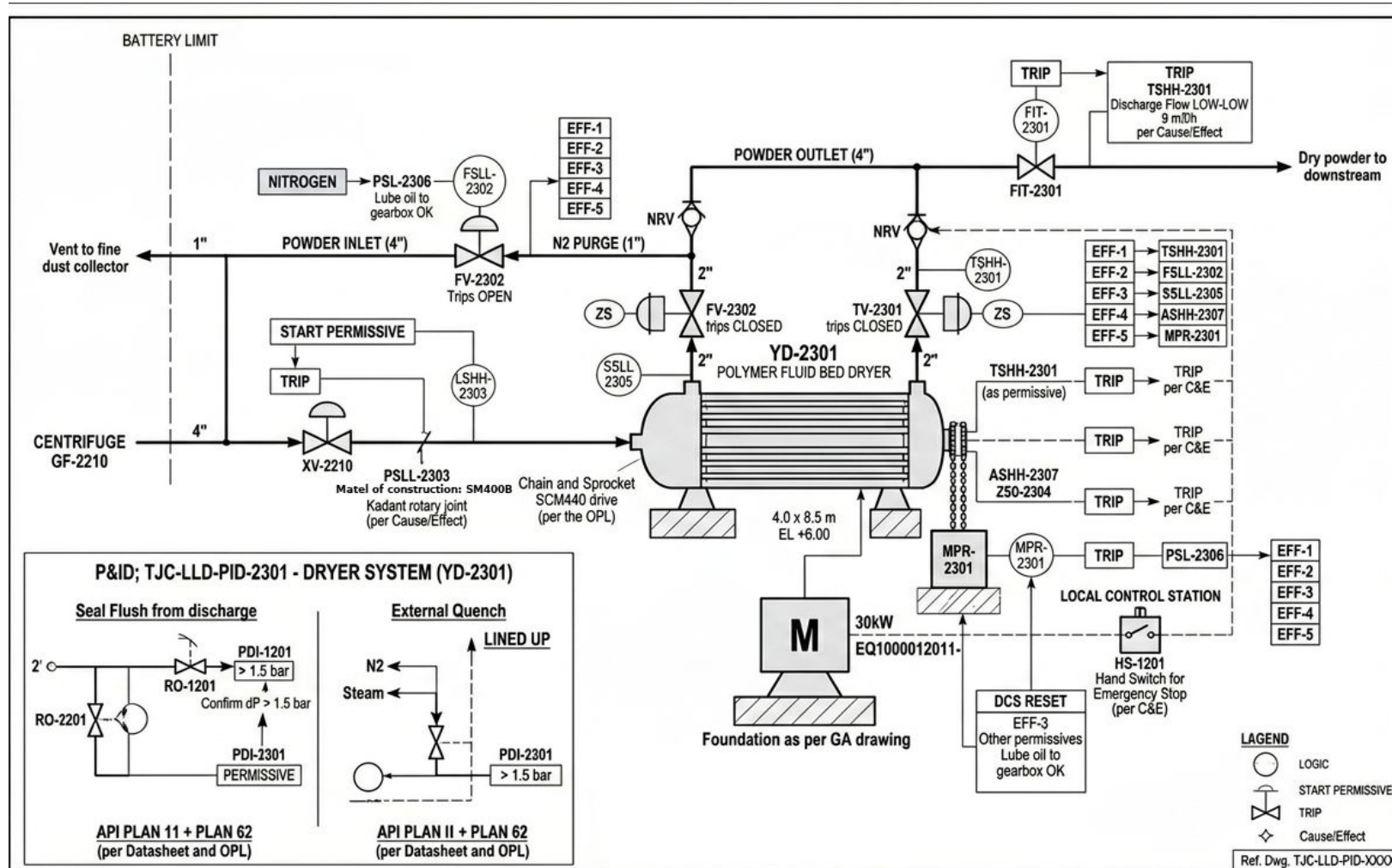
- Feed transfer from storage tank to reactor/process
- Pump protection logic: min-flow bypass, suction pressure, seal flush, vibration, temperature trips

What it can extract

- Pump, valve, line, flow, pressure, motor tags
- Trip, permissive, alarm and seal flush conditions

How it is useful

- Explains start-up, low-flow and suction issues
- Links logic with SOP and maintenance history



NOTE 1: This is sample data provided for CALIBER purposes only.

NOTE 2: Refer to Datasheet TJC-LLD-DS-GA-1201A for design parameters.

NOTE 3: See SEQ-1201 logic diagram

What it shows

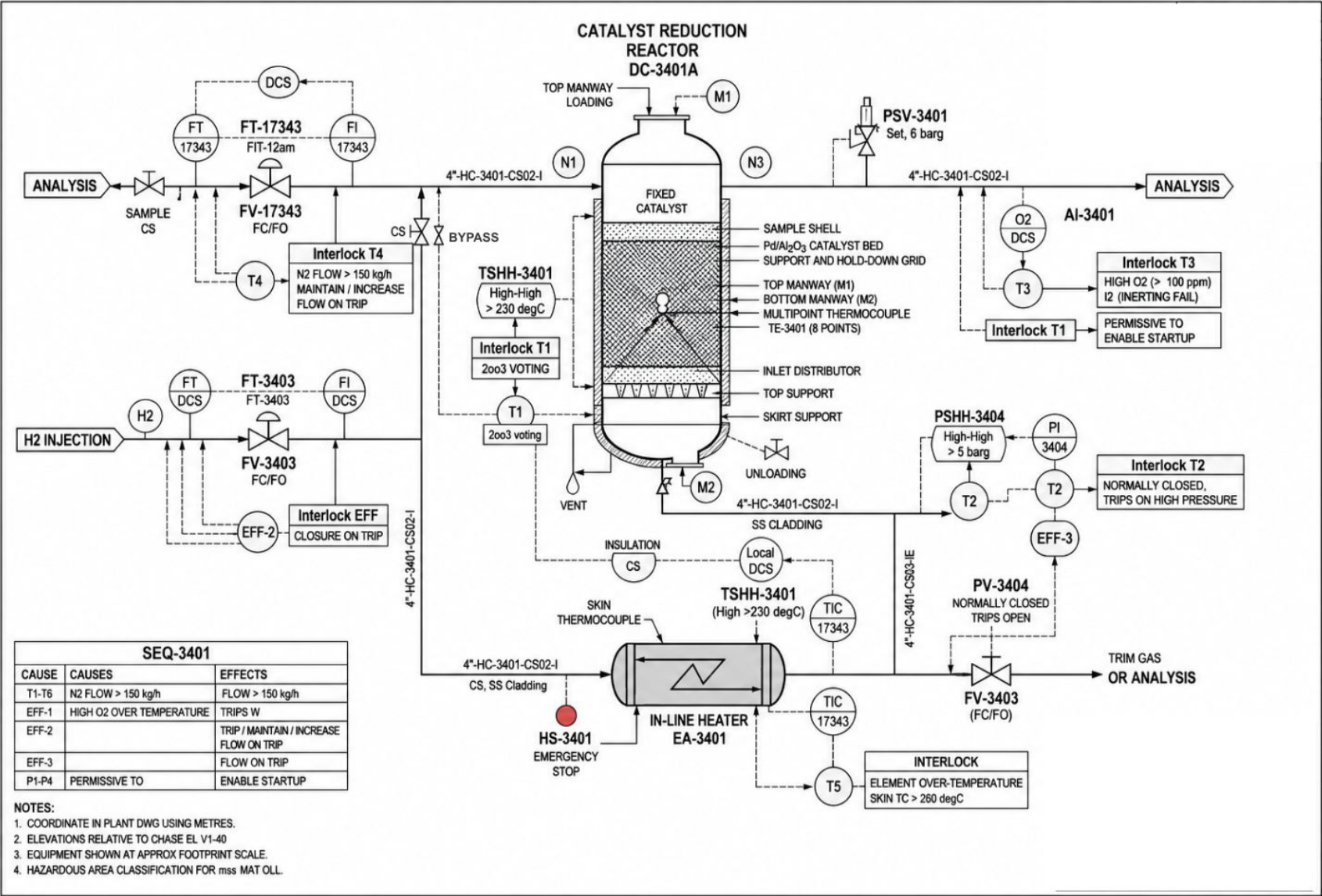
- Dryer with powder inlet/outlet, N2 purge, venting and motor operation
- Permissive and trip logic for flow, pressure, temperature, lube oil, vibration

What it can extract

- Dryer, centrifuge, N2 purge, vent, motor and instrument tags
- Start permissive, ESD, DCS reset and trip logic

How it is useful

- Explains dryer trips and operating requirements
- Connects SOPs, purge requirements and maintenance records



What it shows

- Hydrogen injection, catalyst bed, analysis point and temperature control
- Safety-critical interlocks, permissives and high-temperature protection

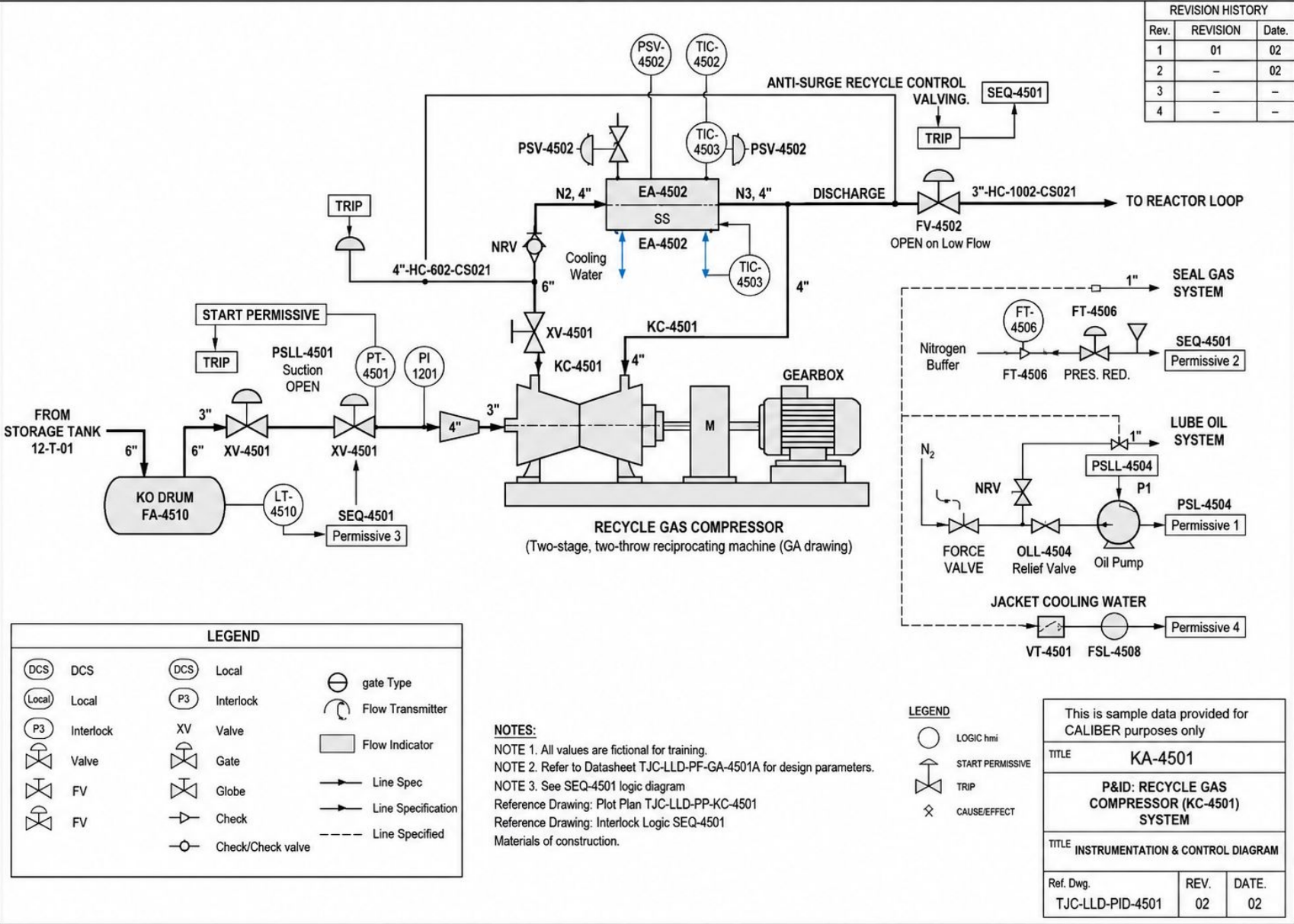
What it can extract

- Reactor, heater, analyzer, PSV, flow, temperature and interlock tags
- Cause-and-effect logic and trip conditions

How it is useful

- Answers safety-critical questions with references
- Supports startup and temperature-interlock troubleshooting

P&ID #SET 4 - Recycle Gas Compressor System



What it shows

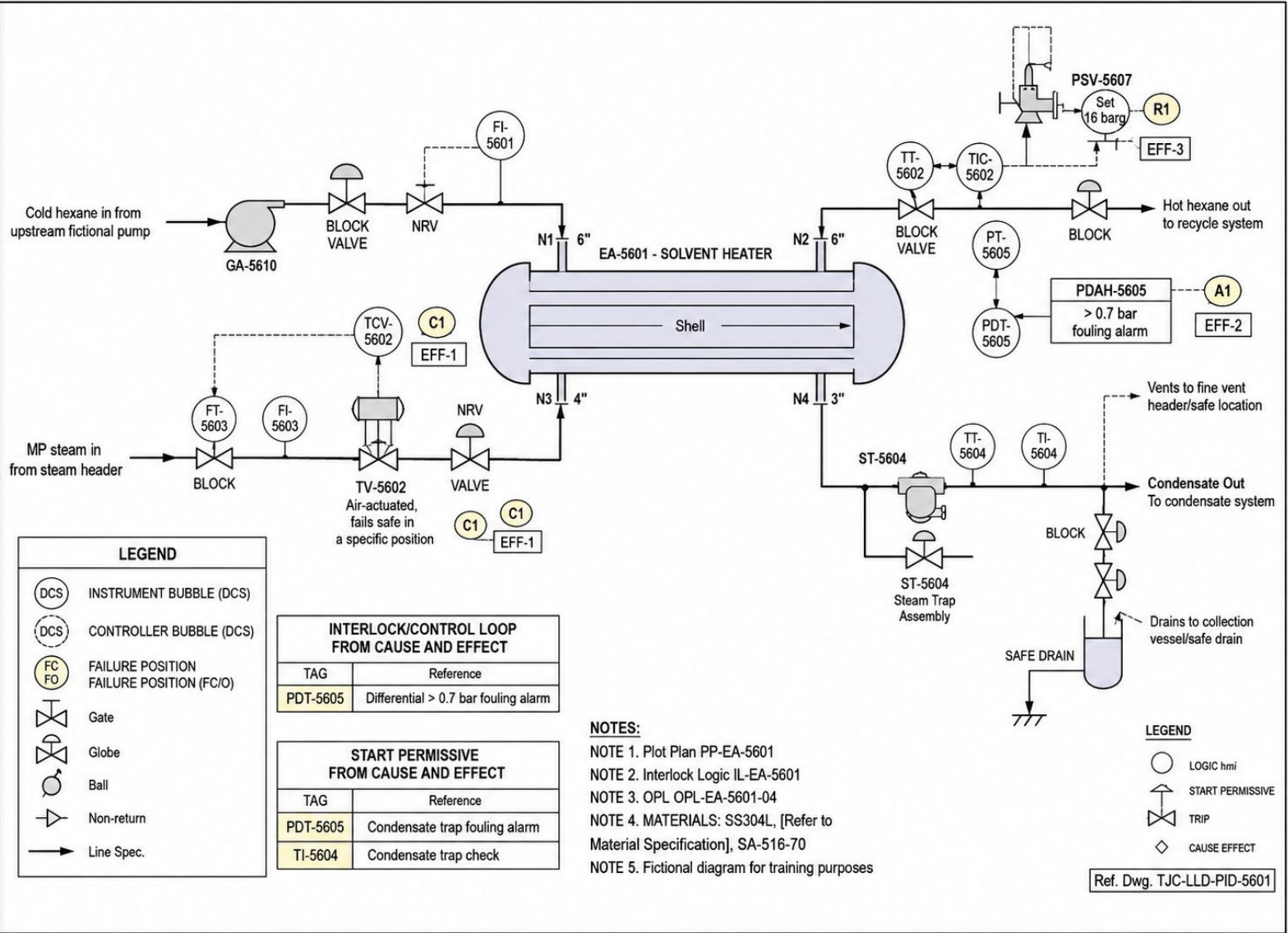
- Recycle compressor with anti-surge protection
- Seal gas, lube oil and cooling-water support systems

What it can extract

- Compressor, recycle valve, trip and permissive tags
- Anti-surge logic and equipment protection data

How it is useful

- AI helps Investigate compressor trips and root causes
- Improve reliability and operational decision-making



What it shows

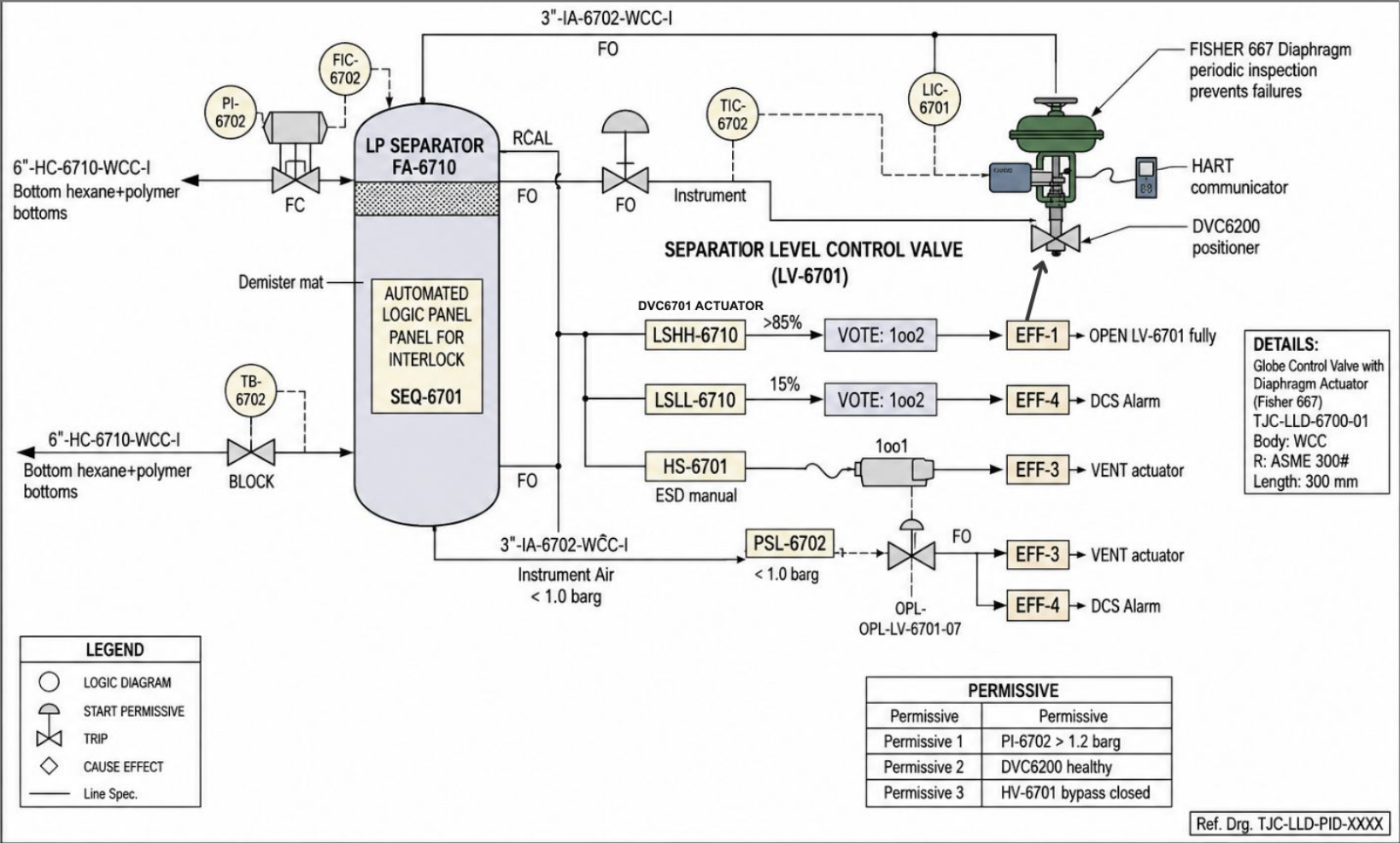
- Hexane heating using steam shell-and-tube heater
- Temperature control, condensate and steam trap system

What it can extract

- Heater, PSV, steam valve, trap and alarm tags
- Fouling alarm and process protection information

How it is useful

- Helps AI support heater performance troubleshooting and fouling analysis.
- Support energy and maintenance optimization



What it shows

- LP Separator and level-control valve system
- Instrument air, HART communication, and voting logic

What it can extract

- Separator, LIC, LV-6701, DVC positioner tags
- High/low level alarms, ESD actions, permissives

How it is useful

- Diagnose level excursions and valve issues
- Supports control valve troubleshooting, alarm interpretation, and safer operating decisions.



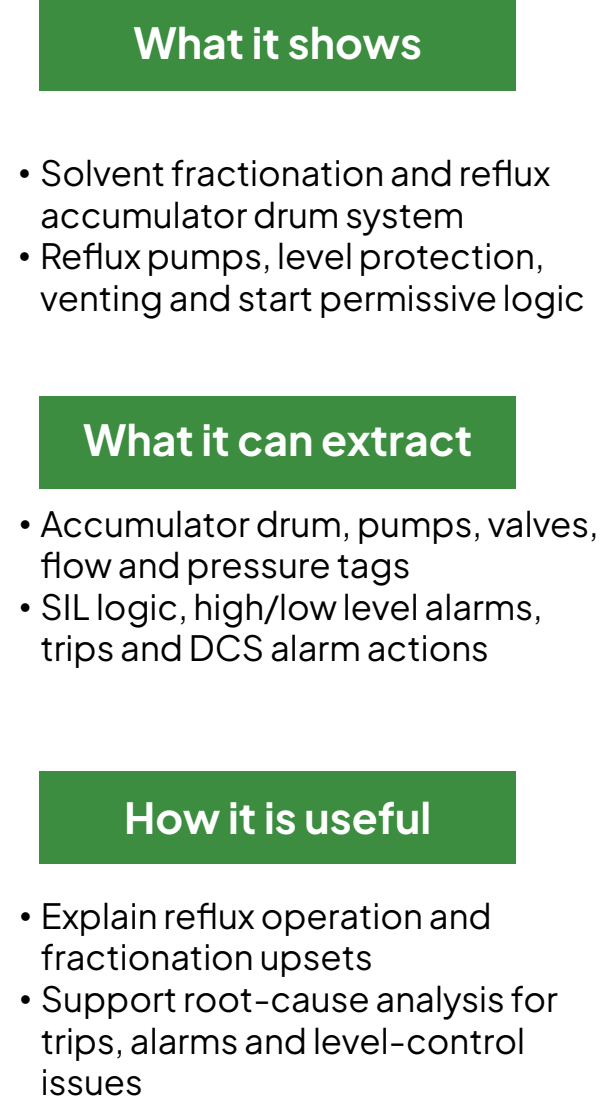
- Cooling tower fan, gearbox, motor and cooling cell operation
- Interlocks related to vibration, oil temperature and motor load

What it can extract

- Fan, gearbox, motor, vibration, temperature and permissive tags
- Trip logic, spare-cell activation logic and DCS notifications

How it is useful

- AI helps analyze fan trips and cooling-system reliability issues
- Assist maintenance planning for vibration, gearbox problems, oil pressure verification, and motor overload investigation.



Maintenance History (Excel)

	1	WO_Number	Notification_No	Report_Date	Start_Date	Completion_Date	Status	Equipment	Equipment Functional	Area_Code	Area_Nam	Plant	Work_Type	Discipline	Priority
2	WO-240109	NT-2024-560424	6/4/2024 8:16	6/5/2024 8:16	6/6/2024 3:16	Completed	KC-4501	RECYLE (TIC-LLD-4)	4500	GAS RECO LINEAR LO	Inspection Instrumen	Medium			
3	WO-240157	NT-2024-560614	6/9/2024 15:36	6/10/2024 16:36	6/11/2024 18:36	Completed	LV-6701	SEPARATO TIC-LLD-6	6700	PRODUCT LINEAR LO	Predictive	Mechanic	Low		
4	WO-240035	NT-2024-560138	6/14/2024 8:53	6/15/2024 7:53	6/15/2024 17:05	Completed	YD-2301	POLYMER TIC-LLD-2	2300	POLYMER LINEAR LO	Calibration Instrumen	Medium			
5	WO-240181	NT-2024-560713	6/14/2024 10:58	6/16/2024 9:58	6/16/2024 18:58	Completed	CT-7801	COOLING TIC-LLD-7	7800	COOLING LINEAR LO	Predictive	Electrical	Low		
6	WO-240087	NT-2024-560339	6/18/2024 6:27	6/19/2024 19:27	6/20/2024 7:27	Completed	KC-4501	RECYLE (TIC-LLD-4)	4500	GAS RECO LINEAR LO	Corrective Instrumen	High			
7	WO-240128	NT-2024-560498	6/20/2024 6:53	6/20/2024 18:53	6/20/2024 23:23	Completed	EA-5601	SOLVENT TIC-LLD-5	5600	SOLVENT LINEAR LO	Preventive Instrumen	Low			
8	WO-240186	NT-2024-560733	6/22/2024 15:06	6/23/2024 20:06	6/24/2024 6:06	Completed	CT-7801	COOLING TIC-LLD-7	7800	COOLING LINEAR LO	Inspection Instrumen	Medium			
9	WO-240085	NT-2024-560332	6/22/2024 15:21	6/25/2024 8:21	6/27/2024 1:21	Completed	KC-4501	RECYLE (TIC-LLD-4)	4500	GAS RECO LINEAR LO	Overhaul	Mechanic	High		
10	WO-240130	NT-2024-560505	6/22/2024 16:00	6/24/2024 2:00	6/26/2024 1:00	Completed	EA-5601	SOLVENT TIC-LLD-5	5600	SOLVENT LINEAR LO	Predictive	Electrical	Low		
11	WO-240195	NT-2024-560767	7/1/2024 9:40	7/1/2024 15:40	7/2/2024 12:40	Completed	FA-8901	REFLUX AT TIC-LLD-8	8900	SOLVENT LINEAR LO	Corrective	Mechanic	Medium		
12	WO-240190	NT-2024-560745	7/4/2024 13:15	7/5/2024 6:15	7/6/2024 16:03	Completed	FA-8901	REFLUX AT TIC-LLD-8	8900	SOLVENT LINEAR LO	Corrective	Mechanic	Medium		
13	WO-240177	NT-2024-560696	7/5/2024 12:51	7/6/2024 16:51	7/7/2024 5:21	Completed	CT-7801	COOLING TIC-LLD-7	7800	COOLING LINEAR LO	Preventive	Mechanic	Low		
14	WO-240103	NT-2024-560404	7/6/2024 15:25	7/7/2024 0:25	7/8/2024 14:25	Completed	KC-4501	RECYLE (TIC-LLD-4)	4500	GAS RECO LINEAR LO	Predictive	Electrical	Low		
15	WO-240056	NT-2024-560212	7/8/2024 11:28	7/9/2024 14:28	7/11/2024 10:28	Completed	DC-3401A	CATALYST TIC-LLD-3	3400	CATALYST LINEAR LO	Corrective	Instrumen	High		
16	WO-240138	NT-2024-560545	7/16/2024 10:36	7/17/2024 8:36	7/18/2024 13:54	Completed	LV-6701	SEPARATO TIC-LLD-6	6700	PRODUCT LINEAR LO	Corrective	Instrumen	Medium		
17	WO-240090	NT-2024-560350	7/16/2024 15:48	7/16/2024 17:48	7/17/2024 9:12	Completed	KC-4501	RECYLE (TIC-LLD-4)	4500	GAS RECO LINEAR LO	Calibration	Instrumen	High		
18	WO-240146	NT-2024-560576	7/20/2024 13:24	7/21/2024 15:24	7/23/2024 16:24	Completed	LV-6701	SEPARATO TIC-LLD-6	6700	PRODUCT LINEAR LO	Preventive	Mechanic	Low		
19	WO-240202	NT-2024-560790	7/21/2024 13:02	7/22/2024 22:02	7/23/2024 3:32	Completed	FA-8901	REFLUX AT TIC-LLD-8	8900	SOLVENT LINEAR LO	Preventive	Mechanic	Low		
20	WO-240172	NT-2024-560673	7/22/2024 8:25	7/22/2024 20:25	7/23/2024 10:25	Completed	CT-7801	COOLING TIC-LLD-7	7800	COOLING LINEAR LO	Preventive	Mechanic	Low		
21	WO-240154	NT-2024-560604	7/22/2024 14:45	7/22/2024 18:45	7/22/2024 20:45	Completed	LV-6701	SEPARATO TIC-LLD-6	6700	PRODUCT LINEAR LO	Predictive	Electrical	Low		
22	WO-240012	NT-2024-560049	7/24/2024 13:56	7/25/2024 0:56	7/26/2024 2:50	Completed	GA-1201A	HEXANE FI TIC-LLD-1	1200	FEED PRE LINEAR LO	Inspection	Instrumen	Medium		
23	WO-240129	NT-2024-560502	7/25/2024 8:19	7/28/2024 1:19	7/29/2024 13:19	Completed	EA-5601	SOLVENT TIC-LLD-5	5600	SOLVENT LINEAR LO	Predictive	Electrical	Low		
24	WO-240049	NT-2024-560184	7/26/2024 15:46	7/28/2024 18:46	7/29/2024 3:46	Completed	YD-2301	POLYMER TIC-LLD-2	2300	POLYMER LINEAR LO	Predictive	Electrical	Low		
25	WO-240201	NT-2024-560786	7/27/2024 9:27	7/28/2024 8:27	7/29/2024 14:57	Completed	FA-8901	REFLUX AT TIC-LLD-8	8900	SOLVENT LINEAR LO	Preventive	Mechanic	Low		
26	WO-240059	NT-2024-560229	7/30/2024 8:23	8/1/2024 21:23	8/2/2024 14:23	Completed	DC-3401A	CATALYST TIC-LLD-3	3400	CATALYST LINEAR LO	Corrective	Instrumen	High		
27	WO-240116	NT-2024-560455	7/31/2024 10:32	8/1/2024 4:32	8/2/2024 20:14	Completed	EA-5601	SOLVENT TIC-LLD-5	5600	SOLVENT LINEAR LO	Inspection	Mechanic	Medium		
28	WO-240060	NT-2024-560235	8/5/2024 6:20	8/7/2024 15:20	8/9/2024 1:20	Completed	DC-3401A	CATALYST TIC-LLD-3	3400	CATALYST LINEAR LO	Inspection	Mechanic	High		
29	WO-240039	NT-2024-560149	8/5/2024 11:31	8/5/2024 17:31	8/8/2024 0:37	Completed	YD-2301	POLYMER TIC-LLD-2	2300	POLYMER LINEAR LO	Corrective	Instrumen	High		
30	WO-240009	NT-2024-560032	8/10/2024 8:01	8/11/2024 23:01	8/14/2024 0:19	Completed	GA-1201A	HEXANE FI TIC-LLD-1	1200	FEED PRE LINEAR LO	Corrective	Instrumen	Medium		

Also please see the explanation of each column in this sheet

What it shows

- Historical maintenance records for plant equipment
- Failures, inspections, preventive, predictive, calibration, and overhaul activities
- Downtime, maintenance costs, root causes, and corrective actions for each work order

What it can extract

- Equipment reliability and failure trends
- Recurring issues, root causes, and breakdown patterns
- Downtime, maintenance cost, and spare part consumption analysis

Equipment-level maintenance history using **Equipment Tag** as the common key across datasets

How it is useful

- Identifies high-risk and high-cost equipment
- Supports root cause and reliability analysis
- Connects maintenance history with operational and engineering data
- Enables predictive maintenance and maintenance knowledge hub use cases